

3	(a)	The internal energy of a substance is the sum of the kinetic energy due to the random motion of the molecules and potential energy due to intermolecular forces of attraction.	A1 A1																
	(b)(i)	Work done by gas = area under CA = $(1 \times 10^5)(20-5) \times 10^{-6}$ = 1.5 J	C1 A1																
	(b)(ii)	<table border="1" data-bbox="388 372 1052 709"> <thead> <tr> <th>section of cycle</th><th>heat supplied to gas / J</th><th>work done on gas / J</th><th>increase in internal energy of gas / J</th></tr> </thead> <tbody> <tr> <td>A → B</td><td>zero</td><td>4.2</td><td>+4.2</td></tr> <tr> <td>B → C</td><td>-8.5</td><td>0</td><td>-8.5</td></tr> <tr> <td>C → A</td><td>+5.8</td><td>-1.5</td><td>+4.3</td></tr> </tbody> </table> Last to be calculated : $\Delta U = Q + W$ In one full cycle, the change in internal energy = 0 J	section of cycle	heat supplied to gas / J	work done on gas / J	increase in internal energy of gas / J	A → B	zero	4.2	+4.2	B → C	-8.5	0	-8.5	C → A	+5.8	-1.5	+4.3	A1 A1 A1
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	(c)	Temperature is a measure of the mean kinetic energy of the molecules of an object, i.e. it is the intrinsic property of one system only, but heat is a measure of energy transferred between two systems. Temperature can be measured using a thermometer directly, but heat cannot be measured using a measuring instrument directly. OR Any other differences based on good physics.	B1 B1																